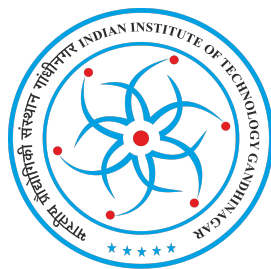


INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

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Subject Name:	Analog IC Design Lab
Subject Code:	EE 653
Assignment No:	06
Submission Date:	20 March 2025
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TASK 1: Designing a Two-Stage OpAmp using Miller Compensation

- Design a 2-stage OPAMP with a Folded Cascode first stage, followed by a Common Source (CS) stage (refer to Fig. 1) for the following specifications. Use the previously designed Voltage and Current Reference for biasing. You need to use Miller Compensation for this task to design the desired circuit.
 - Differential Gain > 100 dB
 - CMRR > 100 dB
 - Output Swing > 1 V
 - UGB > 10 MHz
 - PM $> 65^\circ$
 - Power dissipation < 1 mW
 - $C_L = 1$ pF
- Verify all the specifications of the designed Op-Amp over all the corners (tt, ff, ss, sf, fs).

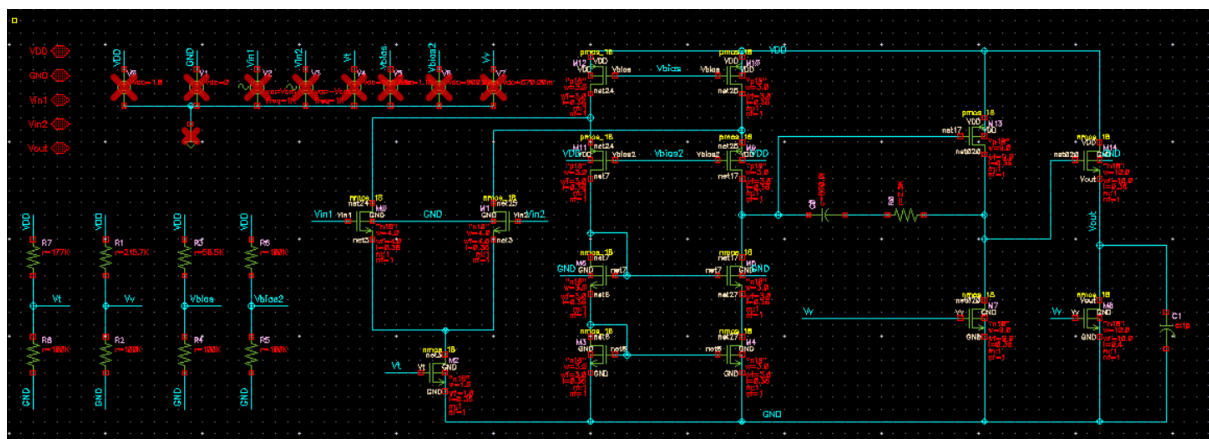


Figure 1: Schematic

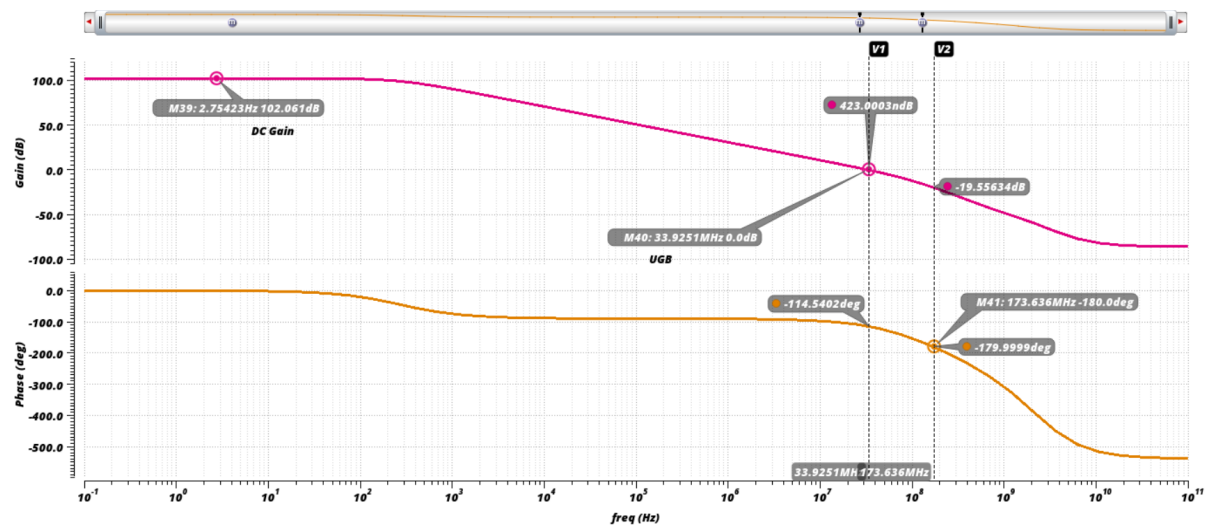


Figure 2: DC Gain

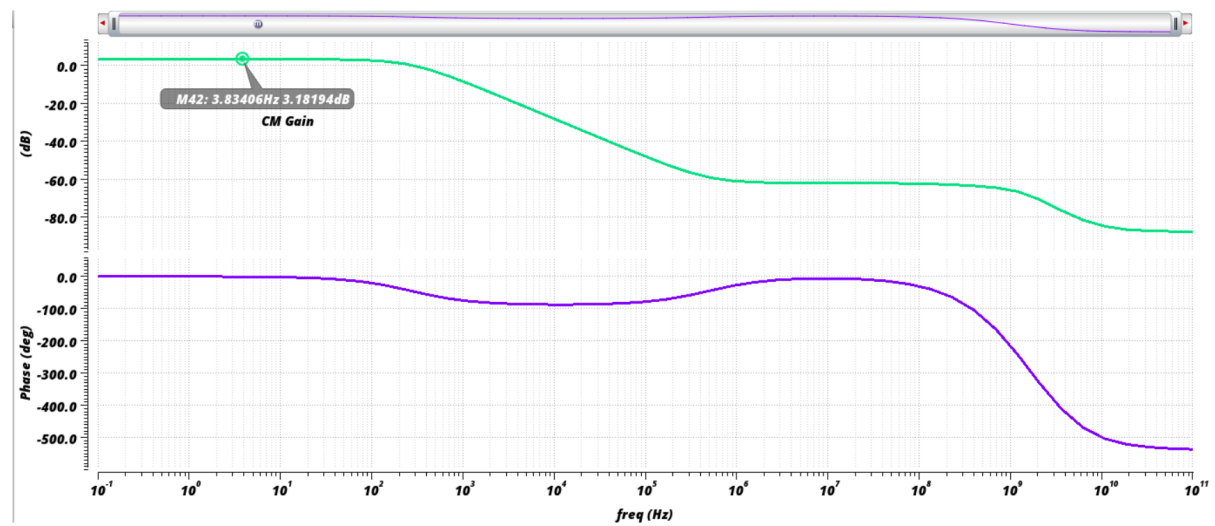


Figure 3: Common Mode Gain

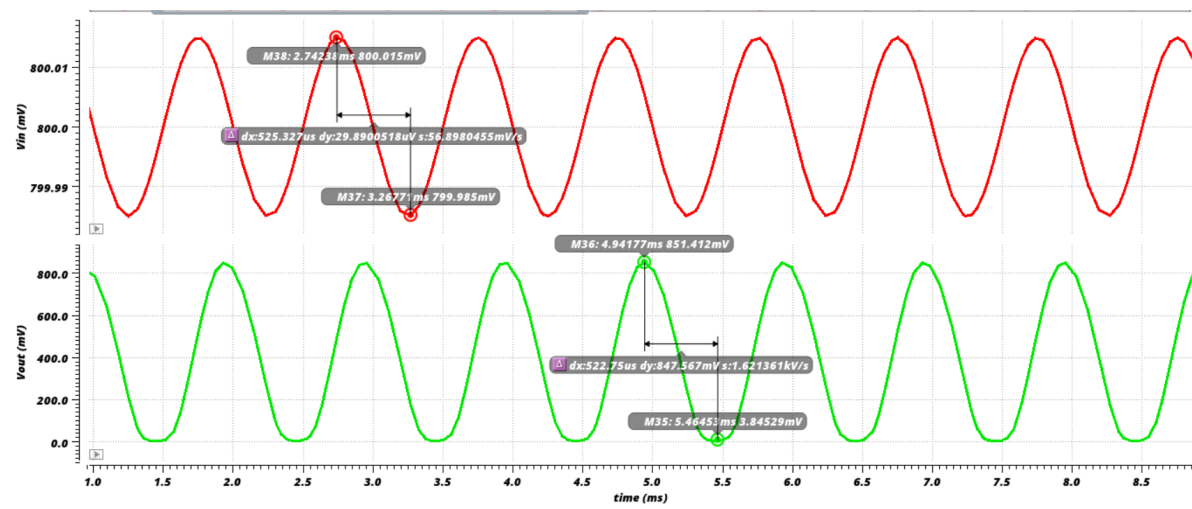


Figure 4: Transient

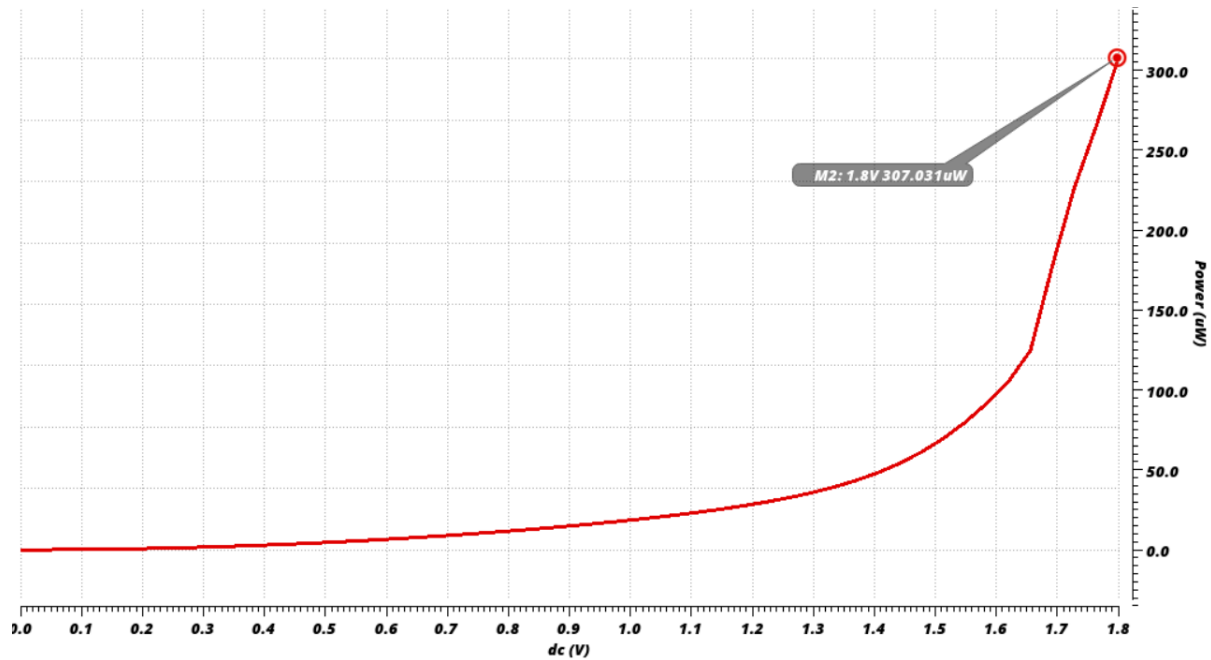


Figure 5: Total Power

Component(s)	Type	W (μm) / Value	L (μm)	Function
M1	NMOS	4.0	0.36	Input differential pair
M2	NMOS	1.0	0.36	Tail current source (bias)
M3, M4	NMOS	3.0	0.36	Current mirror (input stage)
M5, M6	NMOS	3.0	0.36	Folded cascode transistors
M7	NMOS	9.0	0.36	Second stage (common source)
M8, M14	NMOS	10.0	0.36	Output stage transistors
M11, M12	PMOS	3.0	0.36	Active load (folded cascode)
M13	PMOS	9.0	0.36	Second stage load
C_C	Capacitor	800 fF	–	Miller compensation capacitor
R_C	Resistor	2.5 k Ω	–	Nulling resistor (stability improvement)
C_L	Capacitor	1 pF	–	Load capacitor

Table 1: Grouped Component Parameters of the Two-Stage Op-Amp

Device(s)	I_D	V_{ov}	V_{DS}	g_m	g_{ds}	Region
M1	5.785 μA	125.44 mV	275.41 mV	112.70 μS	478.99 nS	Saturation
M2	11.57 μA	182.39 mV	267.39 mV	103.81 μS	468.46 nS	Saturation
M3, M4	5.789 μA	35.38 mV	498.27 mV	103.63 μS	29.43 nS	Saturation
M5, M6	5.789 μA	28.81 mV	592.64 mV	106.06 μS	88.78 nS	Saturation
M11	5.789 μA	140.26 mV	451.85 mV	65.06 μS	926.44 nS	Saturation
M12	11.56 μA	221.23 mV	279.18 mV	90.10 μS	5.19 μS	Saturation
M13	58.35 μA	278.87 mV	784.43 mV	376.19 μS	5.14 μS	Saturation
M7	58.35 μA	110.93 mV	1.015 V	753.53 μS	7.40 μS	Saturation
M8	58.37 μA	113.89 mV	383.23 mV	221.45 μS	1.56 μS	Saturation
M14	58.38 μA	91.11 mV	383.23 mV	788.19 μS	3.98 μS	Saturation

Table 2: DC Operating Point Parameters

Corner	DC Gain (dB)	Phase Margin ($^\circ$)	Gain Margin (dB)	UGB (MHz)
TT	102.061	66.4193	-18.8436	35.3236
FF	95.986	76.0926	-24.3827	38.7633
FS	82.1023	65.5567	-12.7363	27.7775

Table 3: Op-Amp Performance Across Different Corners

Parameter	Value
DC Gain	102.061 dB
Common Mode Gain	3.18 dB
Unity Gain Bandwidth (UGB)	33.9251 MHz
Phase Margin (PM)	65.46 $^\circ$
Power Consumption	307.031 μW
Output Swing (Peak-to-Peak)	847 mV

Table 4: Performance Parameters of the Designed Two-Stage Op-Amp

TASK 2: Designing a Two-Stage OpAmp using Feed-forward Compensation

- Design a 2-stage OPAMP with a Folded Cascode first stage, followed by a Common Source (CS) stage (refer to Fig. 2) for the following specifications. Use the previously designed Voltage and Current Reference for biasing. You need to use Feedforward Compensation for this task to design the desired circuit.
 - Differential Gain > 100 dB
 - CMRR > 100 dB
 - Output Swing > 1 V
 - UGB > 50 MHz
 - PM $> 55^\circ$
 - Power dissipation < 1 mW
 - $C_L = 1$ pF
- Verify all the specifications of the designed Op-Amp over all the corners (tt, ff, ss, sf, fs).

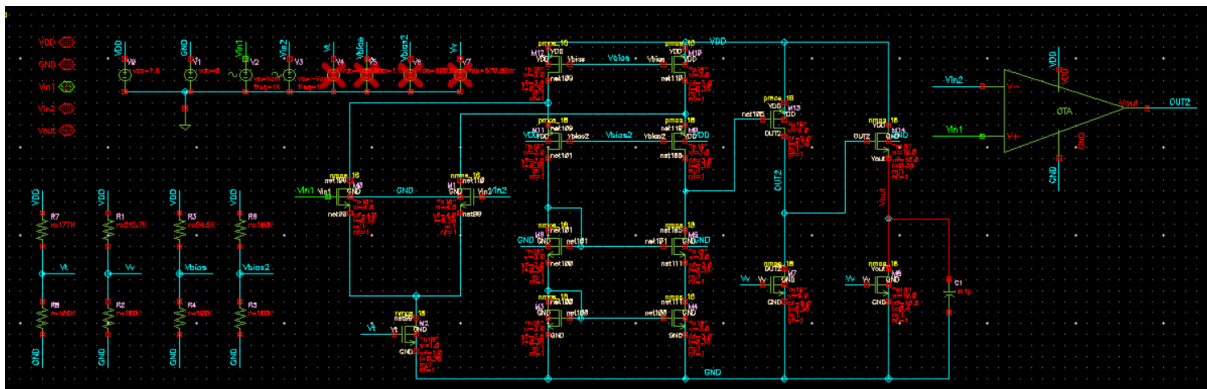


Figure 6: Schematic

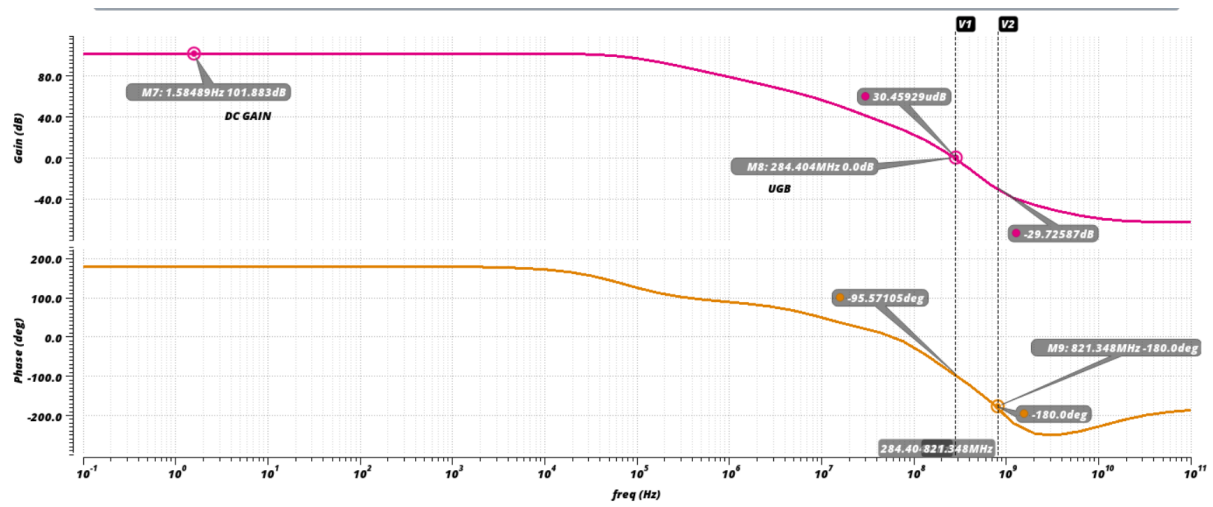


Figure 7: DC Gain

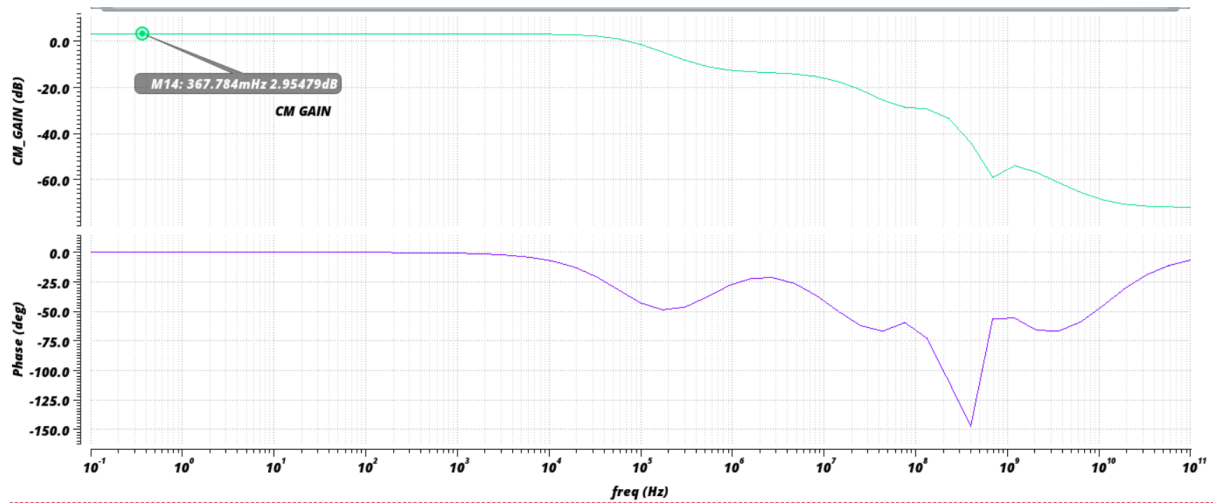


Figure 8: Common Mode Gain

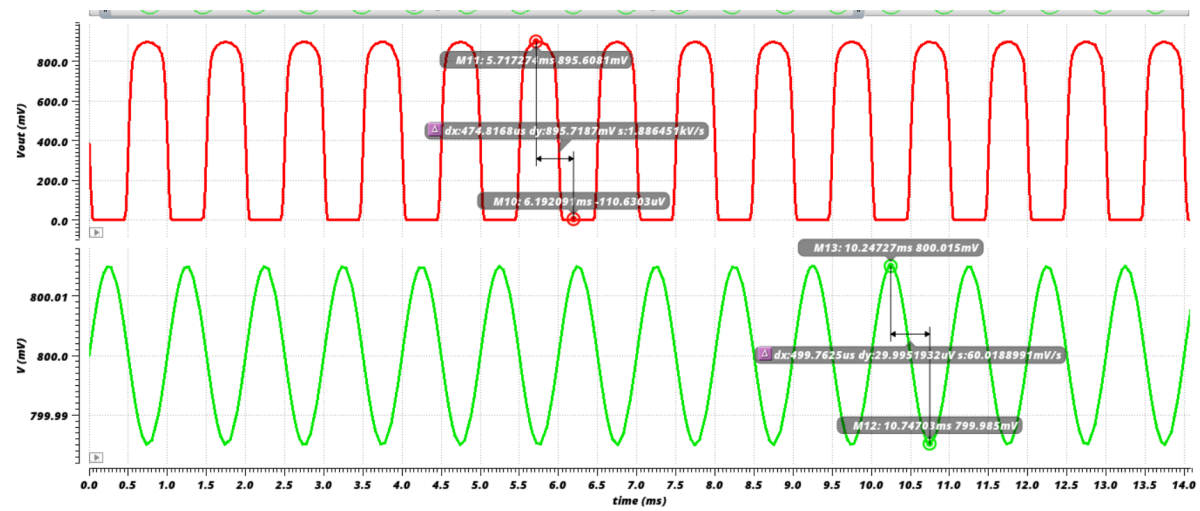


Figure 9: Transient

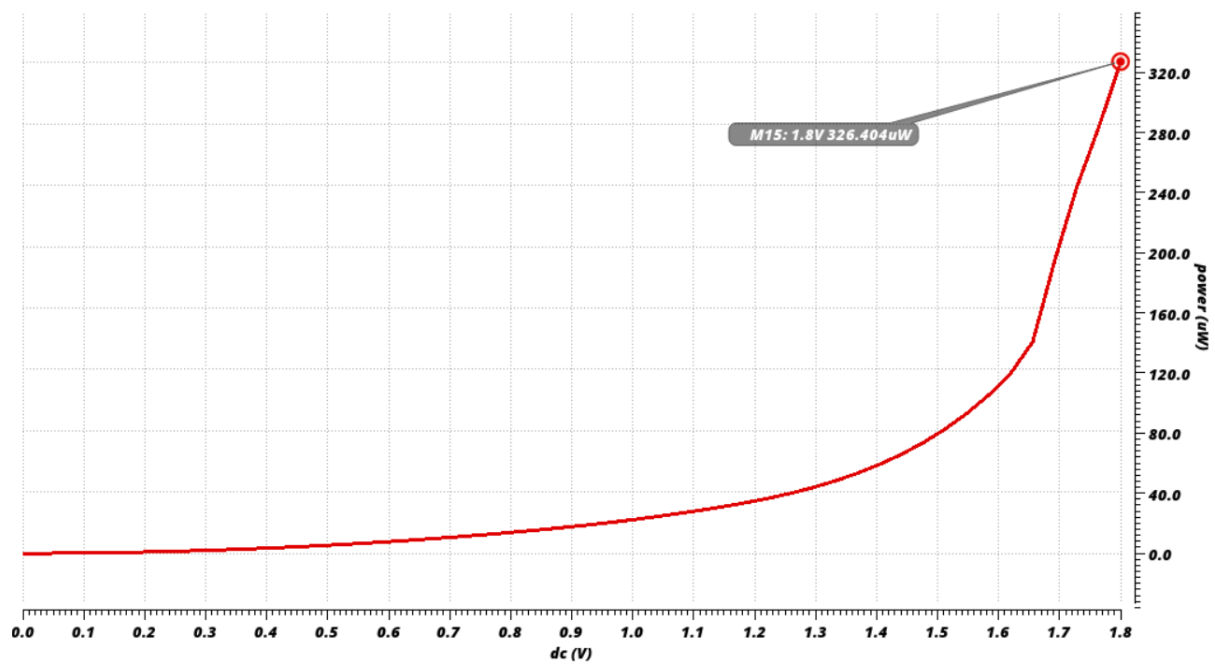


Figure 10: Total Power

Device	Type	W (μm)	L (μm)	Bias/Node
M1	NMOS	4.0	0.36	Input (Vin1)
M2	NMOS	1.0	0.36	Tail current (Vt)
M3	NMOS	3.0	0.36	Current mirror
M4	NMOS	3.0	0.36	Current mirror
M5	NMOS	3.0	0.36	Folded cascode
M6	NMOS	3.0	0.36	Folded cascode
M7	NMOS	9.0	0.36	Second stage
M8	NMOS	10.0	0.40	Output stage
M11	PMOS	3.0	0.36	Active load
M12	PMOS	3.0	0.36	Bias (Vbias)
M13	PMOS	9.0	0.36	Output load
M14	NMOS	10.0	0.36	Output transistor
C_C	Capacitor	800 fF	–	Miller compensation capacitor
R_C	Resistor	2.5 k Ω	–	Nulling resistor (stability improvement)
C_L	Capacitor	1 pF	–	Load capacitor

Table 5: Device Dimensions and Components

Device	I_D	V_{GT}	V_{DS}	g_m	g_{ds}	Region
M1	5.785 μA	125.44 mV	275.41 mV	112.70 μS	478.99 nS	Sat
M2	11.57 μA	182.39 mV	267.39 mV	103.81 μS	468.46 nS	Sat
M3	5.789 μA	35.38 mV	498.27 mV	103.63 μS	29.43 nS	Sat
M4	5.789 μA	35.38 mV	498.27 mV	103.63 μS	29.43 nS	Sat
M5	5.789 μA	28.81 mV	592.64 mV	106.06 μS	88.78 nS	Sat
M6	5.789 μA	28.81 mV	592.64 mV	106.06 μS	88.78 nS	Sat
M11	5.789 μA	140.26 mV	451.85 mV	65.06 μS	926.44 nS	Sat
M12	11.56 μA	221.23 mV	279.18 mV	90.10 μS	5.19 μS	Sat
M13	58.35 μA	278.87 mV	784.43 mV	376.19 μS	5.14 μS	Sat
M7	58.35 μA	110.93 mV	1.015 V	753.53 μS	7.40 μS	Sat
M8	58.37 μA	113.89 mV	383.23 mV	221.45 μS	1.56 μS	Sat
M14	58.38 μA	91.11 mV	383.23 mV	788.19 μS	3.98 μS	Sat

Table 6: DC Operating Point Parameters

Parameter	Value
DC Gain	101.883 dB
Unity Gain Bandwidth (UGB)	284.404 MHz
Phase Margin (PM)	84.43°
Output Swing	895mV
Total Power	326.404uW

Table 7: AC Performance of the Designed Op-Amp

TASK 3: Two-Stage OpAmp using Nested-Miller Compensation

- Design a 2-stage OPAMP with a Folded Cascode first stage, followed by a Common Source (CS) stage (refer to Fig. 3) for the following specifications. Use the previously designed Voltage and Current Reference for biasing. You need to use Nested Miller Compensation to design the desired circuit for this task.
 - Differential Gain > 140 dB
 - CMRR > 140 dB
 - Output Swing > 1 V
 - UGB > 5 MHz
 - PM $> 65^\circ$
 - Power dissipation < 1 mW
 - $C_L = 1$ pF
- Verify all the specifications of the designed Op-Amp over all the corners (tt, ff, ss, sf, fs).

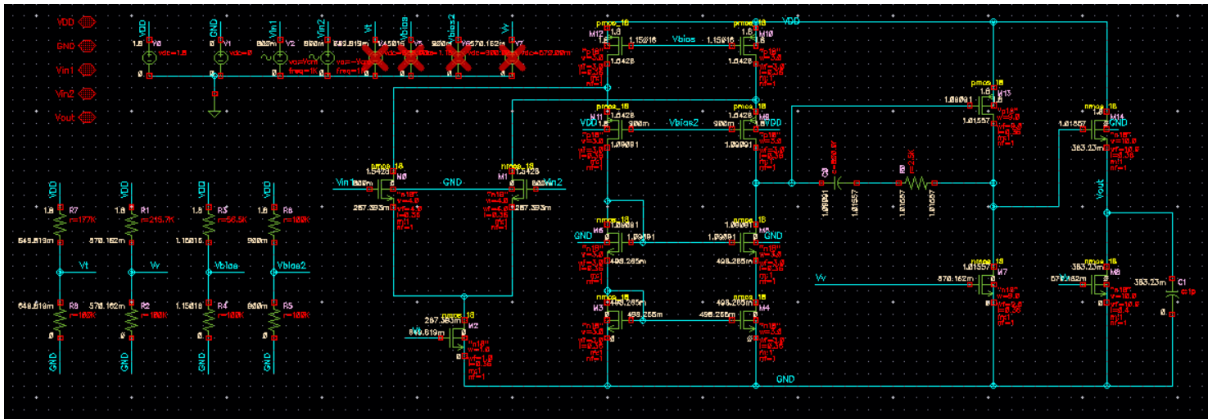


Figure 11: Schematic

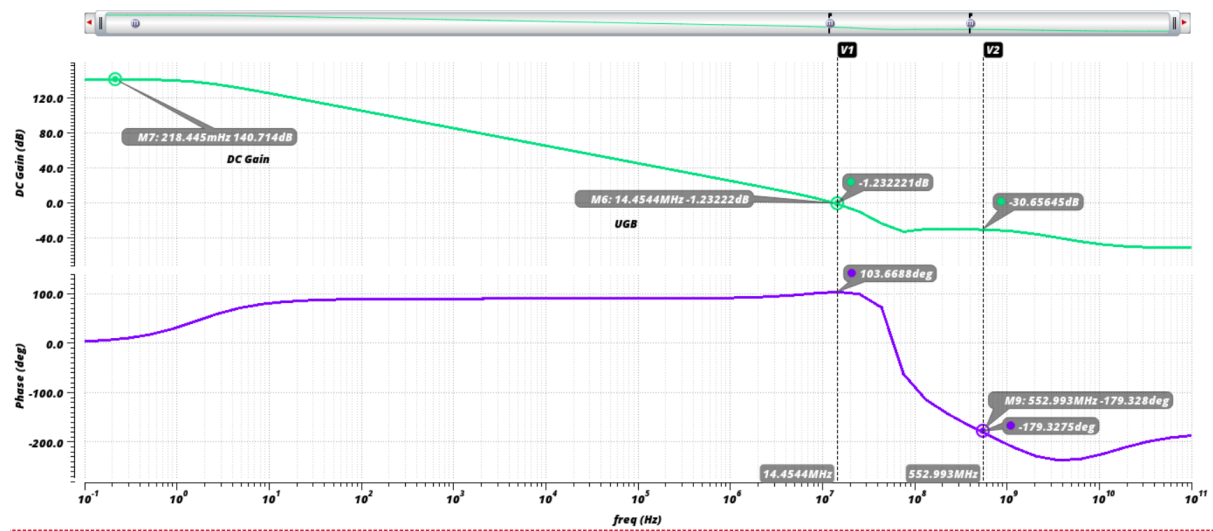


Figure 12: DC Gain

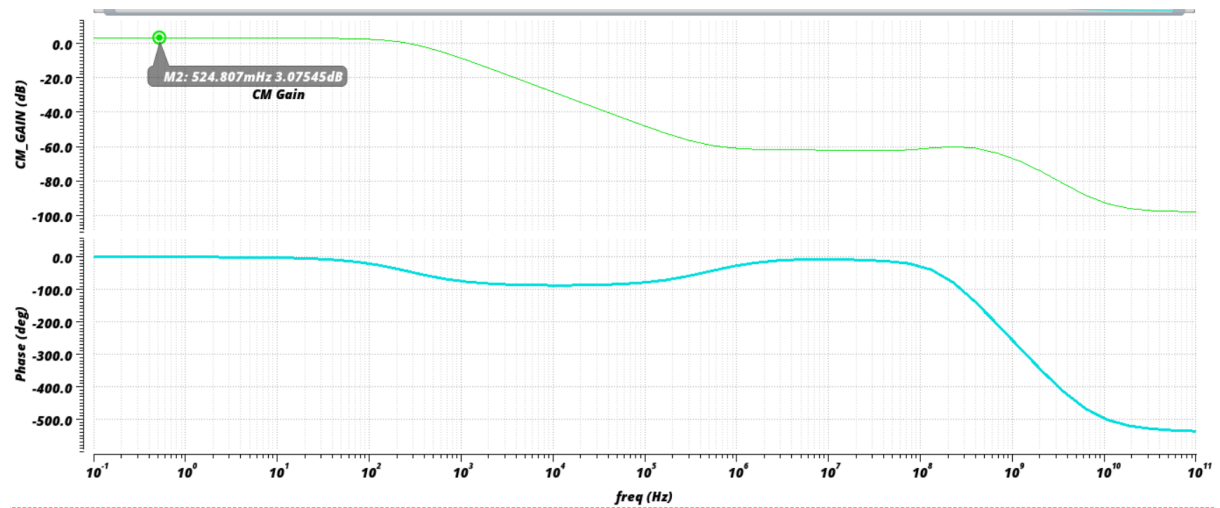


Figure 13: Common Mode Gain

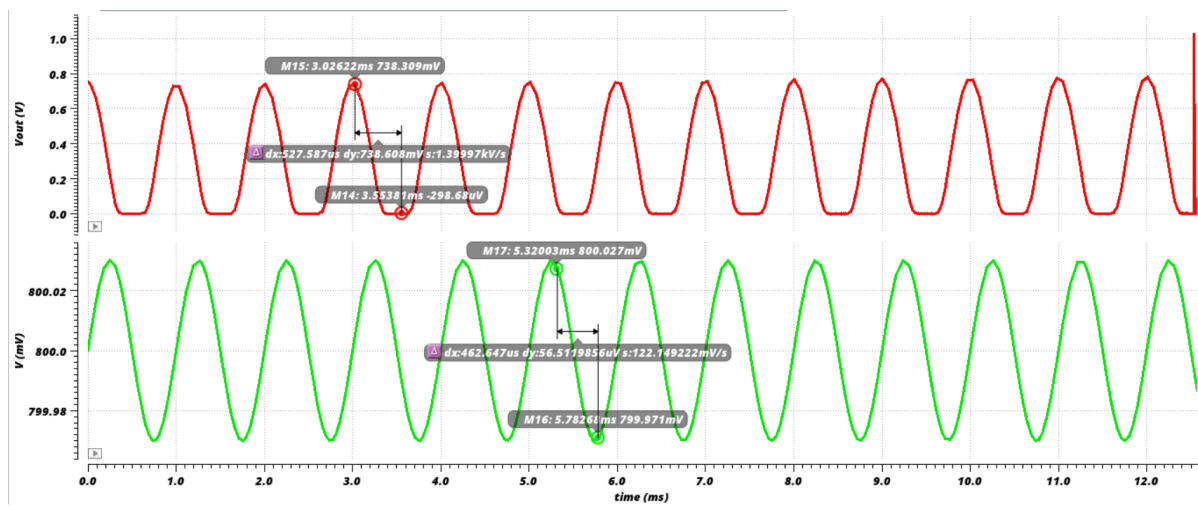


Figure 14: Transient

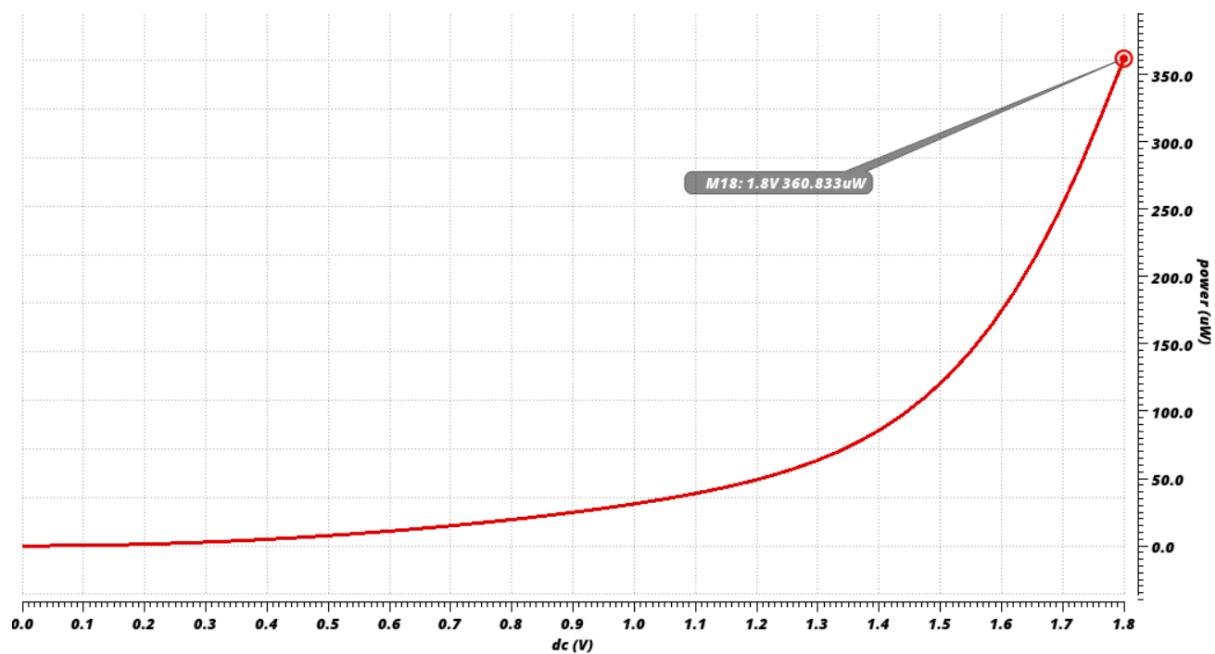


Figure 15: Total Power

Device(s)	Type	W (μm)	L (μm)
M1	NMOS	4.0	0.36
M2	NMOS	1.0	0.36
M3, M4	NMOS	3.0	0.36
M5, M6	NMOS	3.0	0.36
M7	NMOS	9.0	0.36
M8, M14	NMOS	10.0	0.36
M11, M12	PMOS	3.0	0.36
M13	PMOS	9.0	0.36

Table 8: Transistor Dimensions of OP-AMP

Device	I_D	V_{ov}	V_{DS}	g_m	g_{ds}	Region
M1	6.04 μA	13.42 mV	1.10 V	117.43 μS	610.16 nS	Sat
M2	12.09 μA	182.39 mV	440.17 mV	110.10 μS	486.85 nS	Sat
M3, M4	5.537 μA	33.01 mV	498.27 mV	106.19 μS	702.46 nS	Sat
M5, M6	5.537 μA	25.64 mV	582.29 mV	102.48 μS	786.21 nS	Sat
M11	5.58 μA	136.42 mV	485.56 mV	63.44 μS	885.56 nS	Sat
M12	11.58 μA	221.23 mV	286.31 mV	90.34 μS	4.90 μS	Sat
M13	59.53 μA	278.87 mV	552.57 mV	360.86 μS	5.36 μS	Sat
M7	59.53 μA	110.89 mV	1.45 V	756.28 μS	1.68 μS	Sat
M8	58.53 μA	113.9 mV	752.12 mV	751.55 μS	5.22 μS	Sat
M14	58.53 μA	90.78 mV	1.04 V	817.44 μS	5.68 μS	Sat

Table 9: DC Operating Point Parameters Op-Amp

Component(s)	Type	W (μm) / Value	L (μm)	Function
M1	NMOS	3.6	0.36	Tail current source (bias)
M2, M6	NMOS	1.8	0.36	Differential pair (input stage)
M3, M4	PMOS	1.2	0.36	Active load (current mirror)
C_L	Capacitor	200 fF	–	Output capacitor

Table 10: 5T-OTA parameters

Parameter	Value
DC Gain	140.714 dB
Common Mode Gain	3.075 dB
Unity Gain Bandwidth (UGB)	14.4544 MHz
Phase Margin (PM)	65.46°
Power Consumption	360.833 μW
Output Swing	$\approx 738 \text{ mV}$

Table 11: Performance Parameters of Nested Miller Compensated Op-Amp

TASK 4: Comparison

1. Compare the Op-Amps designed using Miller Compensation, Feedforward Compensation, and Nested Miller Compensation for all the specifications over all corners, and comment on the stability of each circuit.
2. Connect all three Op-Amps in unity gain configuration and obtain the step response. Compare the responses by analyzing:
 - Settling time (within 5% of the final value)
 - Peak-to-peak overshoot and undershoot
3. For Task-1 and Task-2 Op-Amps, report the Phase Margins when connected with feedback factors $\beta = 0.2$ and $\beta = 0.5$. Analyze the results and comment on the stability of both circuits.

Parameter	Task 1 (Miller)	Task 2 (Feedforward)	Task 3 (Nested Miller)
DC Gain (dB)	102.06	101.88	140.71
CM Gain (dB)	3.18	2.95	3.07
UGB (MHz)	33.93	284.40	14.45
Phase Margin ($^{\circ}$)	65.46	84.43	65.46
Power (μ W)	307.03	326.40	360.83

Table 12: Comparison of Two-Stage Op-Amps using Different Compensation Techniques